

SACHIN GANPULE

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EDUCATION:

University of Michigan, Ann Arbor

Expected Graduation: May 2028

Master of Applied Data Science (Concentration in AI)

Rutgers University, New Brunswick, NJ

May 2025

Bachelor of Science in Electrical & Computer Engineering

Relevant coursework: Software Engineering, Operating Systems, Data Structures & Algorithms, Linear Systems & Signals.

SKILLS:

Programming: Python • SQL • Bash • Java • C++ • MATLAB

AI/ML: PyTorch • HuggingFace • Transformers • RAG • Vector Databases (FAISS, Chroma) • QLoRA / LoRA • NLP • Explainable AI • Multi-agent systems • LangChain • LangGraph • CrewAI • OpenAI Agents SDK

MLOps: Docker • Kubernetes • Terraform • AWS (Bedrock, Lambda, App Runner, API Gateway, S3, CloudFront, SageMaker) • GitHub Actions • MLflow • LangFuse observability • CI/CD

EXPERIENCE:

AI Engineering Intern

Hawl Technologies

Jan - May 2026

- Built Google Drive MCP integration with OAuth 2.0 to support content extraction from docs, sheets, slides, PDFs, and txt files.
- Extended AI memory Chrome extension to support ChatGLM: end-to-end scraper, event handshake, and docs so users can capture ChatGLM conversations in a unified memory store.
- Currently constructing a highly-scalable, server-side vector search pipeline (Python, pgvector) integrating HNSW indexing to replace legacy TF-IDF, enhancing search relevance and recall.
- Engineering a better production-ready backend for LLM memory storage by developing a batched embedding API, implementing server-side embedding persistence, and migrating rate limiting to Redis for cross-worker reliability.

Data Analytics Extern

Webacy

Aug - Sep 2024

- Performed data validation, correlation analysis, and clustering using Pandas and Matplotlib to profile risks and identify vulnerability patterns for over 1000 smart contracts against forty risk tags.
- Leveraged SQL and Excel to conduct data manipulation and visualization, presenting actionable insights to drive future decision-making in a PowerPoint presentation with an accompanying Tableau dashboard.

PROJECTS:

Multi-Agent Researcher System

Jan 2026 - Current

- Designing a multi-agent pipeline using CrewAI + LangGraph, including planner, researcher, writer, and validator agents.
- Integrating web search, long-context reasoning, and vector storage for persistent memory.

Autonomous Trading Floor

Mar 2026

- Architected a multi-agent system featuring 4 autonomous agents making live market trades powered by 6 MCP servers and 44 specialized tools.
- Deployed the system as an enterprise-grade SaaS Financial Planner using a scalable AWS architecture (Aurora Serverless, Lambda, SQS) with LangFuse for observability.

FrostByte (ODSC x Google Agentic AI Hackathon Project)

Dec 2025

- Developed a multimodal AI agent using Gemini 2.5 Flash that processes fridge images to detect ingredients and generate structured recipe suggestions. Engineered a Planner-Executor workflow that synthesizes visual data with user constraints (dietary, budget) to output optimized strategies, such as "Max Waste Reduction" or "Fastest Prep".

GrAldient

Jan - Jun 2025

- Developed an AI-powered grading tool leveraging OCR to automate quiz and test evaluation from scanned PDFs. Reduced grading time for educators by 40% through a trial test.
- Architected an end-to-end OCR pipeline, transitioning from Tesseract 5 to a fine-tuned TrOCR (Vision Transformer) to boost handwriting recognition accuracy from 90% to 99.5%. Integrated a YOLOv8 model for localized text detection and automated preprocessing on AWS EC2 using Bash scripts.
- Presented GrAldient at the 2025 Rutgers ECE Capstone Expo; ranked in the top 15 teams out of the entire senior class.

CERTIFICATIONS:

AWS Certified Cloud Practitioner: *Sep 2025* | Snowflake GenAI Bootcamp: *June 2025* |